

Linear Algebra

Friday 9 October 2026 · Student handout · Original examples · No solutions

See the structure before calculating

Quick reference

$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. For a triangular matrix, multiply the diagonal entries.

A zero row, or two proportional rows, makes the determinant zero. A zero *diagonal entry* alone is not enough.

Spot a reason, not a long calculation

For each matrix, decide whether its determinant is zero without cofactor expansion. Give a specific structural reason. Also give the value for C.

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 0 & 0 \\ 1 & 4 & -2 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \\ 0 & 1 & 2 \end{pmatrix}$$

$$C = \begin{pmatrix} 2 & -1 & 4 \\ 0 & -3 & 1 \\ 0 & 0 & 5 \end{pmatrix}$$

Matrix	Zero / nonzero (value for C)	Structural reason
A		
B		
C		

A useful counterexample

Do zeros on the diagonal alone force a zero determinant? Use D to test that claim.

$$D = \begin{pmatrix} 0 & 2 \\ 3 & 0 \end{pmatrix}$$

Calculate $\det(D)$, then give a one-sentence conclusion

Check your understanding

What extra condition lets you use the diagonal-product rule? Explain the difference between a **zero row** and a **zero diagonal entry**.

Your explanation

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Track the effect of every row operation

Three operations, three determinant rules

Swap: $R_i \leftrightarrow R_j$ reverses the sign.

Scale: $R_i \leftarrow cR_i$ multiplies the determinant by c .

Replace: $R_i \leftarrow R_i + kR_j$ ($i \neq j$) leaves it unchanged.

Reduce only as far as you need

Find $\det M$ using row operations. Stop when the matrix becomes triangular. Record how each operation affects the determinant.

$$M = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 4 & 3 \end{pmatrix}$$

New matrix / row operation	Effect on the determinant

Determinant of your triangular matrix

Now recover $\det(M)$: account for every sign or scale change

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“Row operations never change the determinant.” Correct the sentence precisely. Why does your final value refer to the original matrix, not just the final one?

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Choose an efficient expansion; check it

Cofactor expansion

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$$N = \begin{pmatrix} 2 & 0 & 3 \\ 1 & -2 & 0 \\ 4 & 0 & 1 \end{pmatrix}$$

First expansion: chosen row / column, terms and calculation

Independent expansion: a different row / column

Why must the two answers agree?

Reason

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$$T = \begin{pmatrix} 4 & 2 & -1 \\ 0 & -1 & 3 \\ 0 & 0 & 2 \end{pmatrix}$$

Start with T . Swap rows 1 and 3 to obtain S . Then apply $R_2 \leftarrow R_2 + 5R_1$ to S to obtain V .

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First expansion: chosen row / column, terms and calculation

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Why must the two answers agree?

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Linear Algebra

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